LESSON 2.11

Add and Subtract Decimals

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can add and subtract decimals by lining up the place values and the decimal points.

WHY THIS MATTERS The space station's main fuel tank reads 128.75 liters. A resupply shuttle delivers 46.8 liters of fuel, but the thrusters used 19.35 liters during docking. What is the fuel level after both events?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	3.75 means 3 ones + 7 tenths + 5 hundredths — it is between 3 and 4
Place value *Spanish: Valor posicional*	What a digit is worth based on where it sits in a number.	In 4.38: the 4 = 4 ones, the 3 = 3 tenths (0.3), the 8 = 8 hundredths (0.08)
Tenths *Spanish: Décimas*	The first spot after the decimal point. It shows parts out of 10.	0.7 = 7/10 — like 7 slices of a pizza cut into 10 equal slices
Hundredths *Spanish: Centésimas*	The second spot after the decimal point. It shows parts out of 100.	0.05 = 5/100 — like 5 pennies out of a dollar (100 pennies)
Annex zeros *Spanish: Agregar ceros*	Adding zeros at the end of a decimal so numbers line up evenly.	15.7 becomes 15.70 so you can subtract 15.70 − 8.36 with aligned place values

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The tank has 128.75 liters. A shuttle adds 46.8 liters. I line up the decimal points and annex a zero: 46.80.

1. I add: 128.75 + 46.80 = 175.55 liters.
2. Now the thrusters use 19.35 liters, so I subtract: 175.55 − 19.35 = 156.20.
3. The final fuel level is 156.20 liters. I check with estimation: about 130 + 47 − 19 = 158, which is close, so my answer is reasonable.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's add 12.45 and 7.8. First, how do we write 7.8 with two decimal places? (7.80.)

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I add _____ to _____, I move _____ to the right on the number line and land on _____.

Word bank Decimal | Place value | Tenths | Hundredths | Annex zeros | decimal | place value | line up | tenths | hundredths | right | left

WATCH OUT!

A common mistake

A common mistake in Add and Subtract Decimals is lining up the last digits (the right edges) of each number instead of lining up the decimal points

LESSON 2.11

Add and Subtract Decimals

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Show 6.5 − 2.8 and 3.25 + 1.5 on the number line



HINT Each label shows an operation — work out the value first, then find that exact spot between the whole numbers.

SHOW YOUR THINKING

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 14.6 + 3.85?

(A) 17.45
(B) 18.35
(C) 18.45
(D) 18.55

HINT Notice the mismatch: 14.6 has one decimal place but 3.85 has two. Lining up the POINTS is the whole trick here.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

3 **Circle the letter of the best answer. Show how you know.**

What is 9.05 + 4.7?

(A) 13.12
(B) 13.57
(C) 13.75
(D) 14.75

HINT Watch the middle zero in 9.05 and the missing hundredths in 4.7 — align the decimal POINTS, never the last digits.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

PART 3 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

What is 20.3 − 7.48?

(A) 12.18
(B) 12.82
(C) 12.92
(D) 13.82

HINT 20.3 stops at the tenths but 7.48 has hundredths — you can't subtract from an empty column until you annex a zero.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

5 Circle the letter of the best answer. Show how you know.

What is 50 − 23.47?

(A) 26.53
(B) 26.63
(C) 27.47
(D) 27.53

HINT 50 is a whole number but 23.47 has hundredths — you'll need to give 50 decimal places before you can subtract.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

PART 4 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — Missing Annex Zero

1. **Problem** Add 8.3 + 2.45
2. **Set up** Lined up the right edges so the 3 sits under the 5
3. **Add** 8.3 + 2.45 = 8.75
4. **Final answer** 8.75

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Check the Set up step. Are the decimal points lined up, or just the last digits on the right?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

SENTENCE STARTER When I add ___ to ___, I move ___ to the right on the number line and land on ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.11

Add and Subtract Decimals

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Compute each decimal sum or difference.

Problem	Answer
5.63 + 2.7	
12.4 − 8.56	
9.08 + 4.925	
25.1 − 13.47	
7.006 + 3.94	

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Round each number to the nearest whole number to estimate the sum or difference. If the estimate equals 10, calculate the exact result before sorting. Sort by whether the actual result is greater than or less than 10.

Groups: Result Greater Than 10 | Result Less Than 10

_____ 7.8 + 3.45

_____ 6.25 + 5.9

_____ 15.3 − 4.7

_____ 8.4 − 2.15

_____ 5.6 + 3.82

_____ 12.1 − 4.95

3 **Write the letter of the matching item on each line.**

Match each expression to its answer.

_____ 8.5 + 3.75

_____ 15.2 − 6.8

_____ 4.06 + 7.94

_____ 20 − 13.45

_____ 9.99 + 0.01

_____ 11.3 − 4.8

(A) 12.25

(B) 8.4

(C) 12

(D) 6.55

(E) 10

(F) 6.5

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A student buys items costing $4.59, $12.75, and $3.80. What is the total cost?

(A) $20.14

(B) $21.04

(C) $21.14

(D) $21.24

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — Subtracting Without a Placeholder

1. **Problem** Subtract 12.5 − 3.45
2. **Set up** Left 12.5 as one decimal place instead of annexing a zero
3. **Subtract** 12.5 − 3.45 = 9.5
4. **Final answer** 9.5

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 2.11

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Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the missing number in each equation.

Equation	Missing Number
4.7 + ___ = 10	
___ − 3.85 = 6.15	
8.06 + ___ = 15	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Tasha's Mistake

1. **Problem** 15.7 − 8.36
2. **Set up** 15.7 − 8.36
3. **Subtract** 15.7 − 8.36 = 7.41
4. **Final answer** 7.41

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A runner's lap times are 2.45 minutes, 2.6 minutes, and 2.38 minutes. Her personal record is 7.5 minutes total. Did she beat her record? Show all your work with decimal alignment and explain.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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